

CSE 227: Graduate Computer Security Autumn 2025

Lectures:

Monday/Wednesday 5:00pm-6:20pm FAH 1450

Instructor:

[Earlence Fernandes](#) Office hours: Tuesdays 1pm to 2pm at CSE 3244

TA:

Andrey Labunets, Office Hours: TBD.

Class Resources:

- Submissions on [HotCRP](#)
- Lecture schedule, readings, and course policies on this web page

Grading:

70%: Course Research Project

15%: Attendance via in-person Quiz

15%: Final Project Presentation and Report

Course Overview

This course is on graduate-level computer security, focusing on the software systems foundations of building secure computer systems. We will read a mix of classic and new papers in the field and will cover a range of topics from trusted computing, operating systems, web/cloud systems, and machine learning. Our goal is to illustrate research challenges and solutions. It is not designed to be a tutorial course, but rather to give students the context to understand current security research and evaluate their interest in the field. The course will examine both the defensive and offensive side of the field. At the conclusion of the course, the students will have the foundation to conduct research in computer security and to apply the latest security research to a particular area of practice.

Note to undergrad and professional MS students: This course is research heavy. It is unlike other courses you've encountered in your career where there are a fixed set of topics with exams and specific skills to learn. Rather, we will be reading papers and discussing/analyzing those papers. Your primary performance will

be based on your ability to formulate and tackle research problems in computer security. A quick way to judge whether taking this course is beneficial for your learning is to read a sample paper and ask yourself, "would I be interested in writing such a paper myself?"

Attendance Considerations

This is an in person class. Please do not come to class or exams if you are sick. I will handle requests for remote attendance on a case-by-case basis and I may not grant all requests.

Schedule

Note: Contents more than 1 week into the future is subject to minor changes.

227_Fall25_Plan : Sheet1

Week/Topic	Dates	Reading 1
1 (Trusted Computing)	Sep 29	Intro
	Oct 1	Bootstrapping Trust in Commodity Computers
	Oct 3	Project teams due in HotCRP at 9 pm PT
2 (Authorization)	Oct 6	Confused Deputy
	Oct 8	Macaroons: Cookies with Contextual Caveats for Deceit
	Oct 9	1-Page proposal due in HotCRP at 9 pm PT
3 (Applications of Authz, IFC)	Oct 13	Practical Data Access Minimization in Trigger-Action Policies
	Oct 15	TaintDroid
4 (Sandboxing in the real world)	Oct 20	Zanzibar: Google's Consistent, Global Authorization System
	Oct 22	Invited Lecture by Tyler Potyondy on Tock
5 (Usable Security)	Oct 27	Why Johnny can't Encrypt
	Oct 29	Project workshopping (mandatory attendance)
6 (CPS)	Nov 3	Pacemakers and Implantable Cardiac Defibrillators: Software Radio Attacks and Zero-Power Defenses
	Nov 5	Experimental security analysis of a modern automobile
7 (Computer vision security)	Nov 10	DL crash course (reading assignment is in the next column)
	Nov 12	Robust Physical-World Attacks on Deep Learning Models
8 (LLM Security)	Nov 17	How are LLMs created? (reading assignment next column)
	Nov 19	Universal and Transferable Adversarial Attacks on Alignment Language Models
9 (Misc Topics)	Nov 24	SatCom Security
	Nov 26	Automatically Detecting Online Deceptive Patterns
10 (Finals)	Dec 1	Part 1
	Dec 3	Part 2
Report	Dec 5	Report is due at 12 noon

Research Project

You will work on projects in groups of 2-4. The goal of the project is to conduct original research in Computer Security. You are encouraged to come up with your own project idea, but we have a few ideas that are well-scoped for a quarter project. Talk with me to scope out a project. The project will require a 1-page project proposal containing: (1) What problem you are tackling; (2) Why that problem is worth tackling; (3) The expected contributions to the science of computer security. This will be due by the second or third week of the quarter. At the end of the quarter, you are expected to turn in a short research paper (max 6 pages; two column format) and give a 15 minute talk. We will have periodic status updates to help you stay on track. The project is 70% of your course grade. I recommend that you take this seriously from the start. I especially value projects that are publication-worthy.

Quiz

We will have up to 2 papers per session. You have to read these in detail and should come prepared to discuss the paper. Your goal is to be able to perform a critical evaluation of the following: (1) what is the problem area of the paper; (2) Why is that problem area important; (3) What is the paper's approach; (4) Did the paper do a good job solving the problem; (5) What is the broader context/related work within which this paper's contributions exist; (6) Your thoughts on future work. As the start of every class, there will be a quiz with 1-2 questions that test your understanding of the paper. Quizzes will be administered through a Google form. You need a smartphone to complete the quiz. This quiz is in-person only. If you miss it, there is no make up option. Please ensure you bring a laptop (or equivalent device) to take the quiz. If you do not have computing equipment, talk to me after the first class. There will not be any quiz for the first class.

By taking this course, you implicitly agree to abide by the UCSD policies on [Integrity of Scholarship](#) and [Student Conduct](#). See the [Academic Integrity Support for Remote Learning](#). University rules on integrity of scholarship and code of conduct are taken seriously and will be enforced.

This course content and structure is inspired by prior versions of this course taught at UCSD and at the following institutions as well: UW Madison, Michigan.



The adversary didn't
follow the rules we made up

A definitive guide to cybersecurity

ORLY?

Robert Hackerman

CSE 227: Graduate Computer Security

Earlence Fernandes

Material adapted from J. Alex Halderman (Michigan), Yoshi Kohno (UW), Franz Roesner (UW),
Amir Rahmati (SBU), Kassem Fawaz (UW Madison)



Today's Session

- Welcome, introductions, course goals & overview
- How to read papers, review them, and present them
- Summary of “Reflections on Trusting Trust”



PhD from Michigan
Postdoc at Univ. of Washington
Prof at UW Madison (2019-2022)
Prof at UC San Diego (2022-now)
Research: building secure systems,
security of machine learning

Cycling fanatic

earlence.com

Goals for the Course

- Learn about foundational ideas in building secure computer systems
- Learn about the latest and greatest in computer security
- Selection of topics
 - Trusted computing environments (SGX, RISC-V, TrustZone)
 - Authorization and Access Control (OAuth, Macaroons, Capabilities)
 - Information Flow Control
 - Machine Learning
 - Adversarial examples, Large Language Model security (e.g., ChatGPT)
 - Application/Domain areas
 - Web/Internet, Internet of Things, Cyber-physical systems, Human-computer interaction

Overview

- This is a graduate-level research course designed to equip you with the skills to do research at the highest levels of quality and impact
- Technically:
 - Experience with research helpful, but not required
 - Experience with operating systems, data structures, algorithms
 - Some basic exposure to computer security will be very helpful (CSE 127)
- Doing well:
 - Be able to read security research papers, and synthesize/internalize their ideas
 - Be able to do well on the research project component (more on this later)

CSE 227: Graduate Computer Security Autumn 2023

Lectures:

Tuesday/Thursday 11:00am-12:20pm WLH 2204

Instructor:

[Earlence Fernandes](#) Office hours: Tuesday 1.30pm-2.30pm at CSE 3244

TA:

Alisha Ukani, Office Hours: Wednesday 11am-12pm CSE 3109.

Class Resources:

- Submissions on [HotCRP](#)
- Lecture schedule, readings, and course policies on this web page

Grading:

80%: Course research project
10%: Paper responses
10%: Paper presentation



cseweb.ucsd.edu/~efernandes/teaching/cse227fa23/cse227.html

Course Overview

This course is on graduate-level computer security, focusing on the software systems foundations of building secure computer systems. We will read a mix of classic and new papers in the field and will cover a range of topics from trusted computing, operating systems, web/cloud systems, and machine learning. Our goal is to illustrate research challenges and solutions. It is not designed to be a tutorial course, but rather to give students the context to understand current security research and evaluate their interest in the field. The course will examine both the defensive and offensive side of the field. At the conclusion of the course, the students will have the foundation to conduct research in computer security and to apply the latest security research to a particular area of practice. Note to undergrad and professional MS students: This course is research heavy. It is unlike other courses you’ve encountered in your career where there are a fixed set of topics with exams and specific skills to learn. Rather, we will be reading papers and discussing/analyzing those papers. Your primary performance will be based on your ability to formulate and tackle research problems in computer security.

Pandemic Considerations

This is an in person class. Please do not come to class or exams if you are sick. I will handle requests for remote attendance on a case-by-case basis and I may not grant all requests.

Schedule

Note: Contents more than 1 week into the future is subject to minor changes.

227_Plan : Sheet1

Week/Topic	Dates	Reading 1	Reading 2	Optional material
1 (no response required)	Sep 28	Introduction	Reflections on Trusting Trust (no response required)	How to read a paper
2 (Trusted Computing)	Oct 3	Saltzer Schroeder Design Principles (no response required)	Bootstrapping Trust in Commodity Computers	This world of ours
	Oct 5	Shielding Applications from an Untrusted Cloud with Haven	Keystone: An Open Framework for Architecting Trusted Execution Environments	Watch: MIT Lecture on Enclaves
	Oct 5	Project teams due		

Grading

- 80% research project
- 10% paper responses
- 10% paper presentation/discussion leading

Grading – 10% Reading Papers & Responses

- Reading is a basic research activity: 3 papers per week on average
 - Varies between 1 and 2 papers per session
- Write a ~400 word critical response (these responses are graded)
 - First para: state the problem being solved, summarize contributions (100 words or less)
 - Additional paras: Evaluate strengths and weaknesses, discussing something you would've done differently, suggest new avenues for future work
- Responses due by 8am before the start of the class! Cannot be late (otherwise you get zero for that day)

Enter information about your submission.

Cancel

Save draft

Title

Put paper title here

Abstract

Markdown styling and LaTeX math supported · [Preview](#)

Put your response here

Authors

List the authors, including email addresses and affiliations.

Your Name and UCSD Email here

1.	Email	Name	Affiliation
2.	Email	Name	Affiliation
3.	Email	Name	Affiliation
4.	Email	Name	Affiliation

Grading – 10% Discussion Leading/Presentation

- Presenting work is a big part of your education as researchers!
 - Goal is to communicate complex ideas effectively
- A student will present a short summary of one paper for the day. I will assign students to papers using poll data
- The summary should contain (completed within 20 minutes):
 - Short recap of principal results
 - Strengths and Weaknesses
 - Discussion points
- Feel free to reuse slides from original authors! – Contact them early!
- If there is any background required for a paper, I will present that material
- Important: Practice the presentation with me privately at least 1-2 days in advance of your actual presentation so that you can get feedback

Course Project

- Major component of your grade! (80%)
 - If you do not pay attention to this, your grade will be bad
- Goal should be original research of publishable quality
- Aim to submit a workshop paper at least
- Teams of 2-4
 - Will vary once the course drop off stabilizes.
 - Start talking to people and form groups
 - If you are unsure of participating in this class, DO NOT agree to be a part of a team and then leave them high and dry if/when you drop

Course Project Process

- **Form project teams by Oct 5 at 6 pm:** 2-4 people in a group. Use class discussion board to find group members. Submit your group to course staff via email.
- **Project proposals due by Oct 12 at 6 pm:** Each group submits (via HotCRP) a 1-page proposal including:
 - Problem definition
 - Motivation
 - Planned approach
 - Evaluation plan
 - Milestones and dates
- **Talk to me if you want project ideas – I have several**

Course Project Process

- Nov 2: Peer project workshopping in class:
 - Come prepared to present a summary of your work done so far, and where you are going (10 mins max – will vary based on number of groups)
 - Highlight points for which you'd like feedback. Everyone in the class will give feedback, including myself. We will use an online tool for this – I will send a link when the time comes
 - Everyone is expected to attend this, present their work, and give feedback to others – This is NOT OPTIONAL

Course Project Process

- **Project Presentations Dec 7:**
 - Everyone must participate!
 - Present your project as if it were accepted to a conference
 - I will be grading your work in real time, so all team members need to be present, and must present some component
- **Final reports due Dec 11, 2 pm:**
 - Submit report to course HotCRP
 - Include a component in the report talking briefly about the contributions of each member in the team

If you want to take things further and submit the work to conferences, I am happy to provide guidance on that!

Course Project Process

- During the quarter, I will meet regularly with each group 1:1
- Goal is to refine the research you are doing and to provide fine-grained feedback

Class Mode

- In-person
- We will figure out any other issues as we go along; be flexible

How to Read a Paper: 3 Pass Approach

- First Pass
 - Carefully read title, abstract, intro
 - Read headings, conclusion
 - Glance at math, and references
 - Glance at graphs
 - Answer the 5 C's
 - Category
 - Context
 - Correctness
 - Contributions
 - Clarity

How to Read a Paper: 3 Pass Approach

- Second Pass
 - Read with more care
 - Ignore proofs
 - Write down key points/questions
 - Look carefully at figures/diagrams
 - Mark relevant references you may need
 - Takes a couple of hours in the beginning: will become faster as you gain experience
 - Should be enough to review a paper

How to Read a Paper: 3 Pass Approach

- Third Pass
 - Virtually reimplement the paper
 - Identify weak and strong points
 - Identify and challenge every assumption
 - How would you present it differently?
 - Write down ideas for future work
- For our course, I expect that you do at least 2 passes. 3 passes is great if you are building on top of the paper for your project

Presenting a paper

- Look for presentation video or slides online; cold-email authors!
- Communicate key ideas
- Don't get bogged down in details
- Structure the talk
- Do at least 2 passes on the paper (3 is better)

Law and Ethics

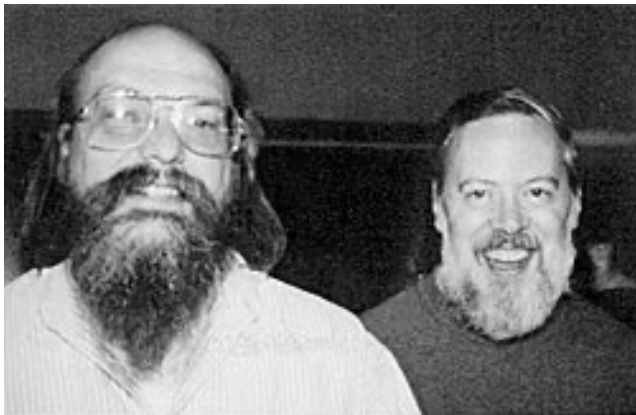
- Don't be evil!
 - Ethics requires you to refrain from doing harm
 - Always respect privacy and property rights
 - Otherwise you will fail this course!
- Federal/state law criminalize computer intrusion/wiretapping
 - E.g., Computer Fraud and Abuse Act (CFAA)
 - You can be sued or end up in jail
- University policies prohibit tampering with campus systems
 - You can be disciplined, even expelled
- Talk to me if you have questions

What this course is not

- An undergrad-like course with a fixed set of topics to teach you (the content of this course varies from year to year, because research moves forward)
- No exams to judge your performance
 - Instead, performance based on your ability to do well at computer security research
 - Reading papers and writing critical responses
 - Conceiving a security research idea and managing it to completion
 - Communicating your ideas effectively and convincing your peers that you solved an important open problem
- No fixed lecturing pattern. Rather, sessions are discussions about papers, their ideas, their merits, their disadvantages, etc.

Reflections on Trusting Trust

1984 Turing Award Lecture



Ken Thompson and Dennis Ritchie,
© WikiQuote

Designed and Implemented Unix

Invented the B Programming Lang
(predecessor of C)

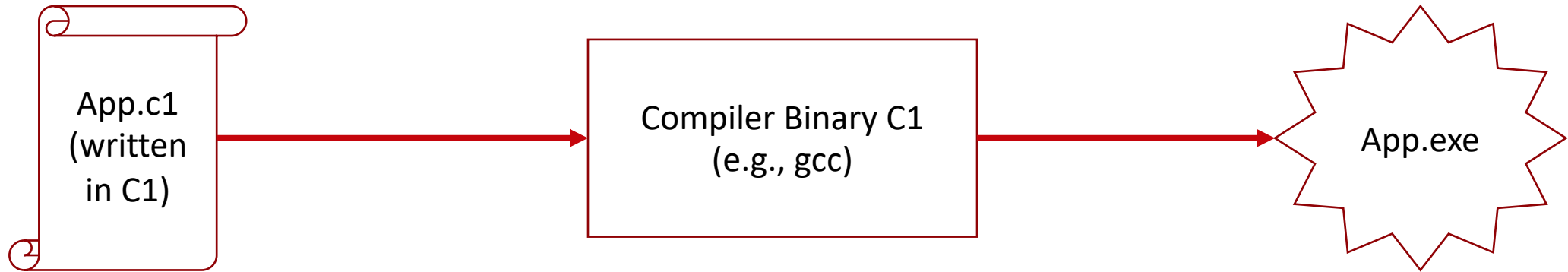
Co-invented Go at Google

Whom and what do you trust when you run software?

- Software developer of the app
- Developer of third-party libraries
- Compiler
- Linker
- Assembler
- Operating System
- Drivers
- External Hardware
- Processor

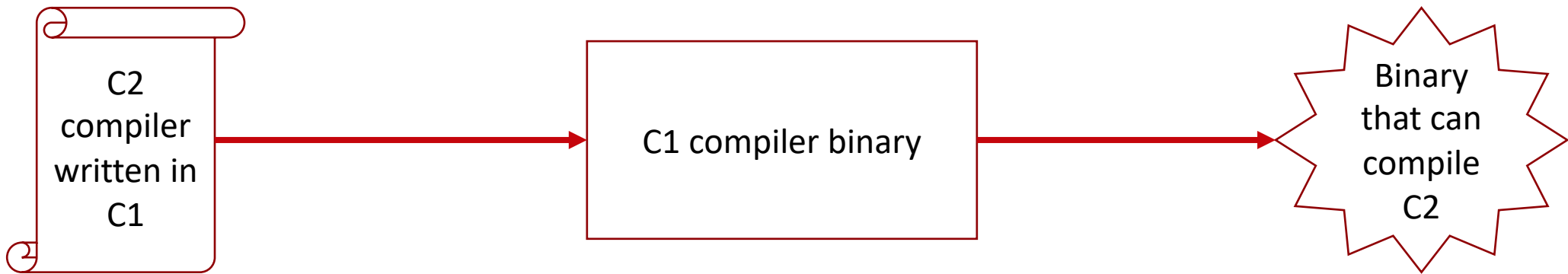


A compiler transforms code from one language to another



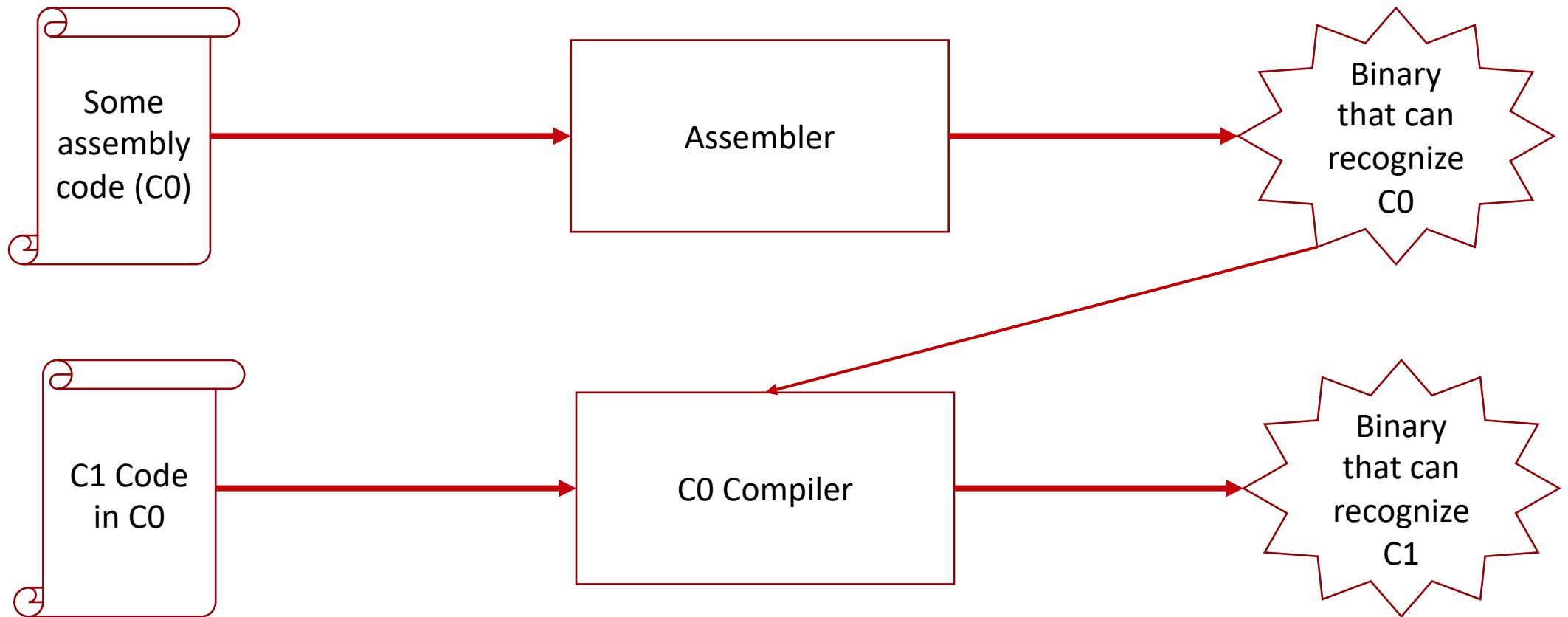
What compiled the compiler?

- Say I have a new version of C1, called C2
 - Write the source of C2 in C1
 - Use compiler binary of C1 to compile C2
 - I get a binary that can compile the C2 language



- But, what compiled the binary that knows how to compile C1?

C0 is a subset of the C1 language

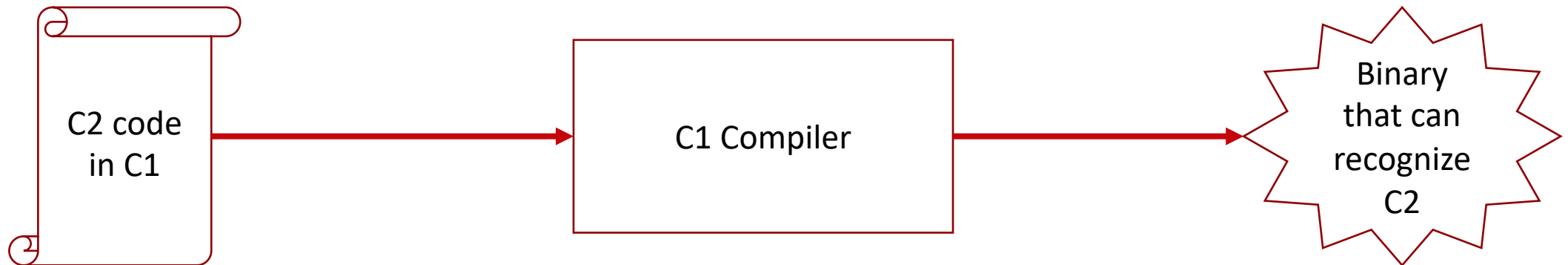


Let's say I want to add a feature to C2

- C2: accepts vertical tab `'\v'` as a special char like `'\n'`, `'\t'`
- Source of C2 written in C1:

```
If (c[0] == '\\ ' && c[1] == 'n')  
    return '\n';  
If (c[0] == '\\ ' && c[1] == 'v')  
    return 11;
```

This is the decimal ASCII
code for vertical tab

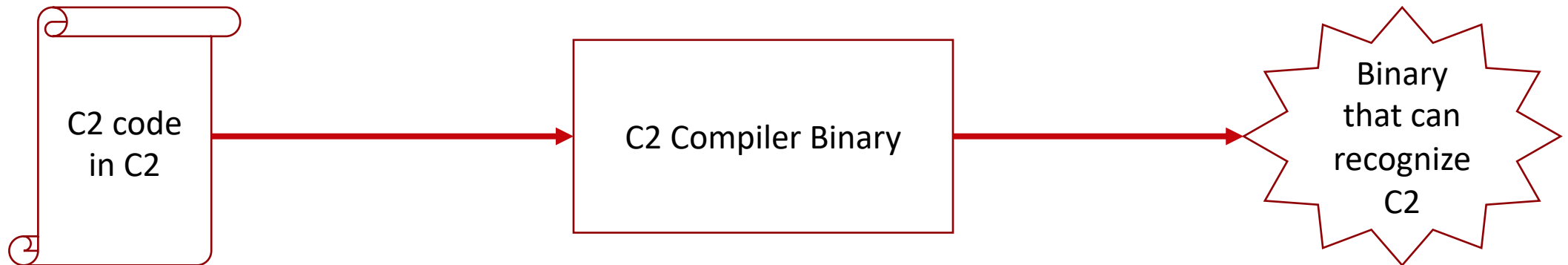


Now C2 can compile itself!

- C2: accepts vertical tab `'\v'` as a special char like `'\n'`, `'\t'`
- Source of C2 written in C2:

```
If (c[0] == '\\' && c[1] == 'n')  
    return '\n';  
If (c[0] == '\\' && c[1] == 'v')  
    return '\v';
```

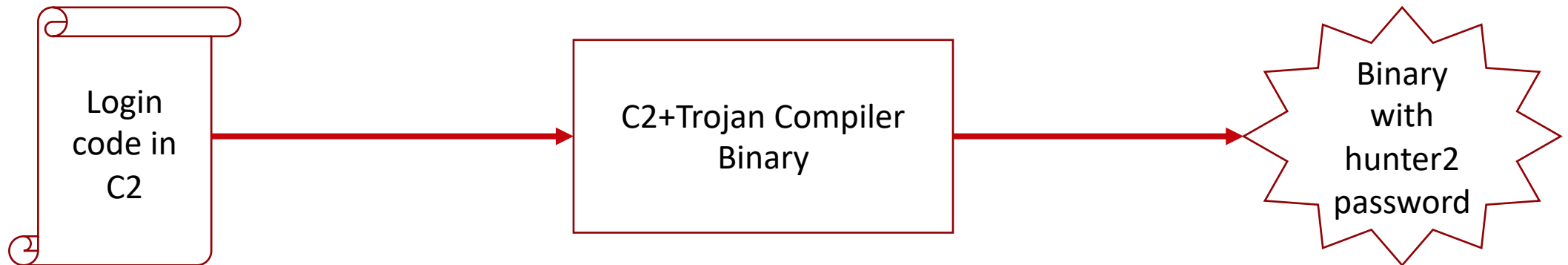
C2 knows what `'\v'` means



Let's modify C2 a bit...

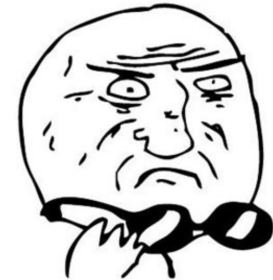
- C2+Trojan: I modify the C2 compiler source in the following way:
 - (recall that compiler is simply processing strings that represent the C2 lang, and then generates instructions)

if (some regex that matches code for a password check)
compile (code that accepts "hunter2" as password in
addition to regular code)

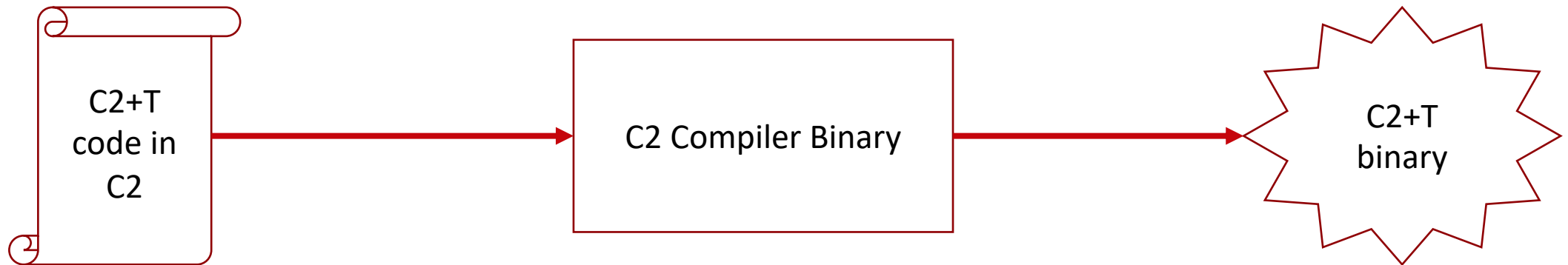


But someone can simply read C2+T source code

if (some regex that matches code for a password check)
compile (code that accepts “hunter2” as password in
addition to regular code)



if (some regex that matches C2 compiler code)
compile (C2 code + hunter2-login code <above>)



Now throwaway C2+T source, install C2+T binary as “official C”

- Then compile the C3 “official source code” (a more complete version of C2)
- There is no backdoor in C3 official source code
- But our C2+T binary will be able to inject the backdoor into the C3 compiler binary
- **THUS, somewhere along the chain-of-trust, there is this malicious binary that can insert backdoors**
- This could be anywhere. Ken Thompson remarks that it could be in the microcode of a processor!

Is any of this real? StrawHorse

[\[edit\]](#) (S//NF) Strawhorse: Attacking the MacOS and iOS Software Development Kit

(S) Presenter: [REDACTED], Sandia National Laboratories

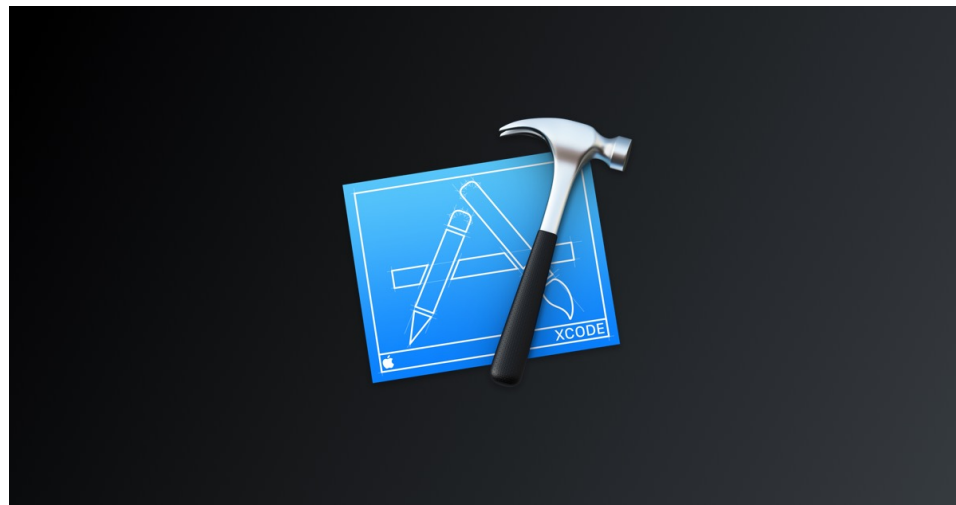
(S//NF) Ken Thompson's gcc attack (described in his 1984 Turing award acceptance speech) motivates the StrawMan work: what can be done of benefit to the US Intelligence Community (IC) if one can make an arbitrary modification to a system compiler or Software Development Kit (SDK)? A (whacked) SDK can provide a subtle injection vector onto standalone developer networks, or it can modify any binary compiled by that SDK. In the past, we have watermarked binaries for attribution, used binaries as an exfiltration mechanism, and inserted Trojans into compiled binaries.

(S//NF) In this talk, we discuss our explorations of the Xcode (4.1) SDK. Xcode is used to compile MacOS X applications and kernel extensions as well as iOS applications. We describe how we use (our whacked) Xcode to do the following things: -Entice all MacOS applications to create a remote backdoor on execution -Modify a dynamic dependency of securityd to load our own library - which rewrites securityd so that no prompt appears when exporting a developer's private key -Embed the developer's private key in all iOS applications -Force all iOS applications to send embedded data to a listening post -Convince all (new) kernel extensions to disable ASLR

(S//NF) We also describe how we modified both the MacOS X updater to install an extra kernel extension (a keylogger) and the Xcode installer to include our SDK whacks.

Is any of this real? XCodeGhost

- Chinese iOS developers downloaded Xcode IDE from some random cloud service because Apple China too slow
- **Xcode was actually XCodeGhost**: modified version to infest the iOS apps compiled
- Collect personal data, phish for credentials (apple ID)



#244321 +(38955)- [X]

<Cthon98> hey, if you type in your pw, it will show as stars

<Cthon98> ***** see!

<AzureDiamond> hunter2

<AzureDiamond> doesnt look like stars to me

<Cthon98> <AzureDiamond> *****

<Cthon98> thats what I see

<AzureDiamond> oh, really?

<Cthon98> Absolutely

<AzureDiamond> you can go hunter2 my hunter2-ing
hunter2

<AzureDiamond> haha, does that look funny to you?

<Cthon98> lol, yes. See, when YOU type hunter2, it shows to us as *****

<AzureDiamond> thats neat, I didnt know IRC did that

<Cthon98> yep, no matter how many times you type hunter2, it will show to us as *****

<AzureDiamond> awesome!

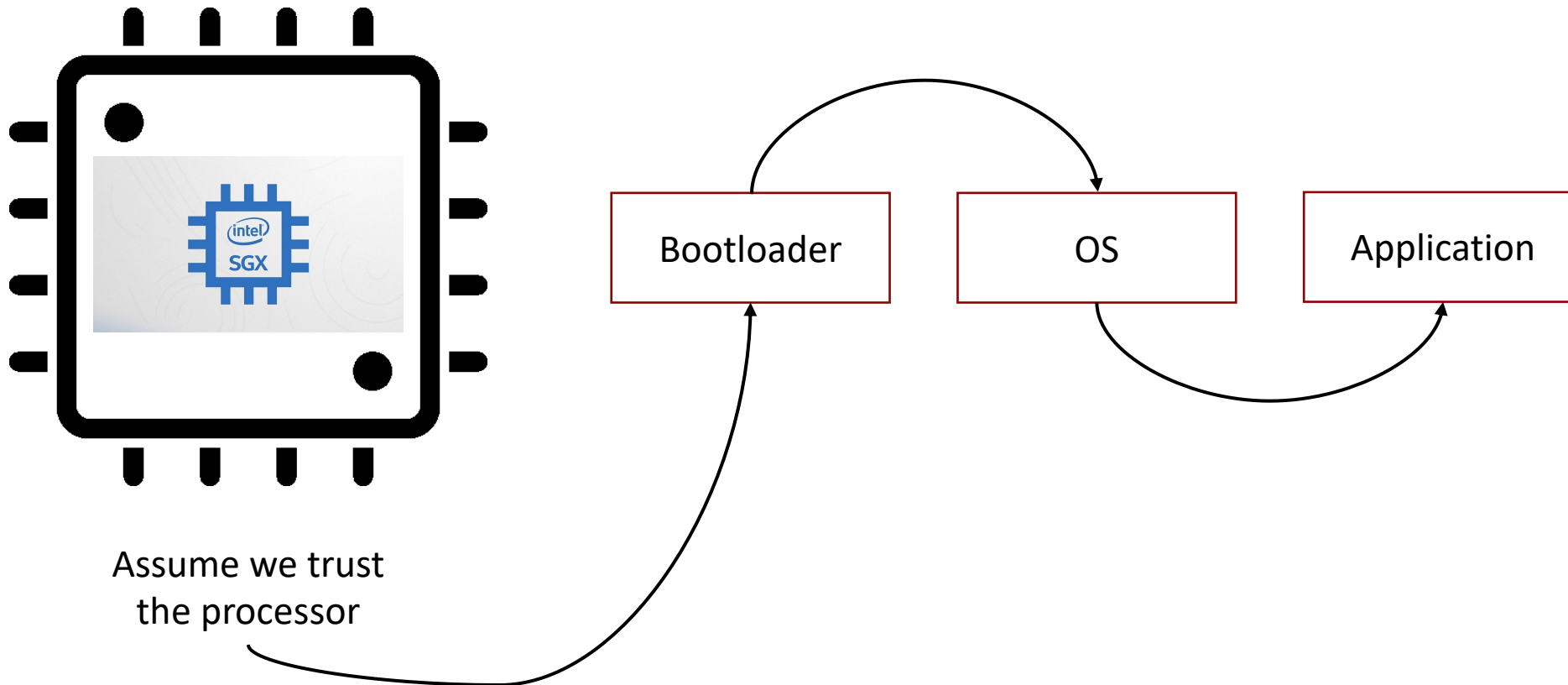
<AzureDiamond> wait, how do you know my pw?

<Cthon98> er, I just copy pasted YOUR *****'s and it appears to YOU as hunter2 cause its your pw

<AzureDiamond> oh, ok.

Root of Trust

Start from something that you trust, and bootstrap trust to higher layers: “chain of trust”



Your Assignments

- First paper reviews due next class (Tuesday)
 - See course site for instructions and submission details
- Submit paper presentation preferences
 - I will email a link today for polling
- Start thinking about your course project
 - Form a group by Oct 5, use Canvas
 - Talk to me for project ideas
- I am out of town on Oct 3
 - No change for you
 - Alisha will be handling the session that day

Financial Aid Survey #bureaucracy

- There is an un-graded quiz that you need to complete inside Canvas
 - More of a survey really
- Federally-mandated thing
- If you don't complete by the end of second week of classes, you will lose fin aid